

The extra-special Steklov extremizer

A matching cover and a three-dimensional comparison

EULER

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1 The original conjecture

Conjecture 1.3 of Lin and Zhao [1]. Let $b \geq 2$ and $r \geq 1$ be integers. Among trees with b leaves and matching number $br + 2$, the extra-special tree $\text{ES}_{b;2r}$ uniquely maximizes the first positive Steklov eigenvalue.

The trees are finite, simple and unweighted; their boundary is the set of leaves, with counting measure. For $b \geq 3$, the source defines $\text{ES}_{b;2r}$ as the spider with arm lengths

$$2r + 2, \quad 2r + 1, \quad \underbrace{2r, \dots, 2r}_{b-2 \text{ arms}}.$$

Arm lengths count edges. The source computes this candidate's eigenvalue, but its conjecture compares it with *all* branching topologies and all leaf counts. The earlier EULER result proves the three-leaf case. Here we prove the full leaf-count statement. The source's definition of ES starts at $b = 3$; at $b = 2$ we state the endpoint explicitly: the extremizer is the path with $4r + 3$ edges, agreeing with the same two-arm recipe.

Put

$$c_b = \frac{3(b-1) + \sqrt{b^2 - 2b + 9}}{2b}.$$

This is the larger root of $bz^2 - 3(b-1)z + 2(b-2)$, and $1 < c_b < 2$. The source value for $\sigma_2(\text{ES}_{b;2r})$ is $(2r + c_b)^{-1}$.

Theorem 1. *Let T have $b \geq 2$ leaves. If its matching number is at least $br + 2$, where $r \geq 1$ is an integer, then*

$$\sigma_2(T) \leq \frac{1}{2r + c_b}.$$

Equality holds exactly for the spider with the displayed arm lengths when $b \geq 3$, and for the path with $4r + 3$ edges when $b = 2$.

The theorem includes the original equality-constrained conjecture and strengthens it to a matching threshold. After its proof, we give a sharp refinement describing the optimum for every possible split of the leaves across the central edge. The argument is uniform in both parameters; it is not an extrapolation from small leaf counts.

2 Use currents instead of boundary potentials

The Steklov matrix Λ is the Schur complement of the interior block of the combinatorial Laplacian. On the space of zero-sum leaf vectors it is positive definite. Write

$$L(T) = \frac{1}{\sigma_2(T)} = \max_{\sum x_i = 0, \sum x_i^2 = 1} x^\top \Lambda^+ x.$$

The inverse form has a particularly simple meaning. Inject current x_i at leaf i . Removing an edge separates the leaves into two sets; conservation forces the current on that edge to be the sum of the injections on either side, up to sign. These currents integrate to potentials because the graph is a tree. Summing their squared values gives the dissipated energy, namely $x^\top \Lambda^+ x$.

In particular, inject $1/\sqrt{2}$ and $-1/\sqrt{2}$ at two leaves and zero elsewhere. Only their joining path carries current. If their distance is d , the energy is $d/2$, so

$$L(T) \geq \frac{1}{2} \text{diam}(T). \quad (1)$$

This makes the matching constraint useful: it will force a long diameter, with a rigid structure in the only length that can compete.

For the rest of the proof put $p = 2r$. A two-leaf tree is a path; matching number at least $2r + 2$ means at least $4r + 3$ edges. Its first positive Steklov eigenvalue is twice the reciprocal of its edge count. This proves that case, so assume $b \geq 3$.

3 A saturated matching cover reveals the whole tree

First the diameter is at least $2p + 3$. Otherwise root the tree at a center, so every depth is at most $p + 1 = 2r + 1$. The vertices at even depths form a vertex cover. There are at most b vertices at each positive depth, since their descendant leaf sets are nonempty and disjoint. The cover thus has at most $1 + br$ vertices, contradicting the matching threshold $br + 2$.

If the diameter is at least $2p + 4$, inequality (1) already gives $L(T) \geq p + 2 > p + c_b$. We need only consider diameter exactly $2p + 3$. Its center is an edge. Delete this edge and root the two halves at its endpoints. Each half has depth at most $p + 1$. These standard center facts follow directly from a diameter path: a branch attached at position j can have length at most $\min\{j, D - j\}$, since neither endpoint can be farther than the diameter D .

Take both roots and every vertex at depths $2, 4, \dots, p$ in the two halves. This is a vertex cover of size at most $br + 2$. Every matching is bounded by every vertex cover, so the assumed threshold forces equality throughout. The matching number is exactly $br + 2$, and every even level in each half has exactly as many vertices as that half has leaves.

The level-two equality is especially strong. Every leaf descends from a level-two vertex, and each such vertex has exactly one descendant leaf. No branching is possible below that level. The level- p equality says that all leaves have depth at least p . Their depths are therefore either p or $p + 1$.

Thus each root is joined by one edge to several *groups*. A group consists of a vertex followed by internally disjoint leaf paths of lengths $p - 1$ or p . Call their leaves short or long, respectively. Single-leaf groups are allowed; the description does not require a group vertex to be a branch vertex.

Saturation of the cover imposes two further restrictions:

- Each group contains at most one short leaf.
- Each half contains a group with no short leaf.

Indeed, a matching of size equal to the cover must saturate every cover vertex and cannot use an edge joining two cover vertices. In particular it cannot use the central edge. A short arm, excluding its group vertex, has r cover vertices but only $r - 1$ other vertices; saturating them forces use of the edge to its group vertex. Hence two short arms cannot share a group. Each root must itself be matched to a group vertex, so at least one group on each side must be free of short arms. This includes $r = 1$, when a short arm has just one edge.

This description covers every possible branching topology that has not already been excluded by the diameter bound.

4 A comparison whose eigenvector has the right signs

Let the two halves contain s and t leaves, so $s + t = b$, and write U for the first set. The current calculation gives, for every zero-sum leaf vector,

$$x^\top \Lambda^+ x = p \sum_i x_i^2 + \sum_{\text{groups } G} \left(\sum_{i \in G} x_i \right)^2 - \sum_{\text{short } i} x_i^2 + \left(\sum_{i \in U} x_i \right)^2. \quad (2)$$

Choose one long leaf a in an all-long group of the first half and one such leaf d in the second. Consider the simpler form

$$F(x) = x_a^2 + x_d^2 + \left(\sum_{i \in U} x_i \right)^2.$$

If the coordinates have one sign on each half, expanding a group square produces only nonnegative cross terms. The subtraction in (2) removes just the short-leaf squares. Therefore the excess over $p \sum x_i^2$ is at least $F(x)$ for such vectors. For a vector with no zero coordinates, equality requires every group to be a singleton and a, d to be the only long leaves.

It is essential to use a maximizing vector of F that has these signs. Both its eigenvalue and the sign property come from a three-dimensional calculation. Let P be the orthogonal projection onto the zero-sum subspace. The nonzero eigenvalues of the compression of F are those of the Gram matrix of $Pe_a, -Pe_d, P\mathbf{1}_U$:

$$\frac{1}{b} \begin{pmatrix} b-1 & 1 & t \\ 1 & b-1 & s \\ t & s & st \end{pmatrix}. \quad (3)$$

Call its largest eigenvalue $\lambda(s, t)$. If $s = 1$ or $t = 1$, its characteristic polynomial is

$$z \left(z^2 - \frac{3(b-1)}{b} z + \frac{2(b-2)}{b} \right),$$

so its largest root is c_b . For a general split, expansion of the same determinant at c_b gives

$$\det(c_b I - \text{matrix (3)}) = -\frac{(s-1)(t-1)}{b} (c_b - 1)^2.$$

When both sides have at least two leaves, this is negative, whereas the monic characteristic polynomial is positive for large arguments. It has a root above c_b . Consequently

$$\lambda(s, t) \geq c_b > 1, \quad \text{with equality exactly when } \min(s, t) = 1. \quad (4)$$

Now take a top eigenvector x of the compressed form F . The projected eigenvalue equation says that on the first half

$$(\lambda - 1)x_a = A, \quad \lambda x_i = A \quad (i \neq a),$$

for one constant A ; on the other half it gives the same equations with d and another constant. Since $\lambda > 1$, all coordinates on each half have one sign. Neither constant can be zero: otherwise the zero-sum condition forces both halves to vanish. Thus every coordinate is nonzero, and the signs on the two halves are opposite. This is exactly the sign condition needed above.

Normalize this vector. Equations (2)–(4) now give

$$L(T) \geq p + F(x) = p + \lambda(s, t) \geq p + c_b.$$

The conjectured inequality follows.

5 Equality and a sharp refinement

Equality in the conjectured bound forces one half to contain a single leaf, every group to be a singleton, and precisely one long leaf on each side. Viewed from the root of the larger half, the arm through the central edge has length $p + 2$; one other arm has length $p + 1$ and the rest have length p . This is exactly the extra-special spider.

Conversely, this spider has matching number $br + 2$. With its center unused the arms have a matching of size $br + 1$; matching the center into its unique odd arm adds one, and no matching can add more than one. Its inverse current form is $p \sum x_i^2 + 2x_a^2 + x_d^2$, precisely the preceding form with a singleton half. Its largest eigenvalue is $p + c_b$. This proves attainability and unique equality in Theorem 1.

The same argument also solves the whole central-split problem, rather than just comparing it with the winning singleton split.

Theorem 2 (Sharp central-split refinement). *Fix an integer $r \geq 1$ and positive integers s, t , and put $b = s + t$. Among trees with matching number at least $br + 2$, diameter $4r + 3$, and s, t leaves on the two sides of the center edge, the maximum first positive Steklov eigenvalue is*

$$\frac{1}{2r + \lambda(s, t)},$$

where $\lambda(s, t)$ is the largest eigenvalue of (3). The unique equality tree has a central edge, one arm of length $2r + 1$ at each endpoint, and all its other arms of length $2r$; there are s arms on one side and t on the other.

Proof. The matching cover forces exactly the group structure already proved. Its same-sign comparison gives $L(T) \geq p + \lambda(s, t)$. Equality forces singleton groups with precisely two long leaves, one per side, which is the displayed tree. For that tree the excess current form is exactly F for every zero-sum vector, so the spectral bound is attained. The matching cover is saturated by matching each center along its long arm and pairing the remaining arm paths; its matching number is $br + 2$. Its two long arms joined by the central edge have length $2p + 3$, so it has the required diameter. These facts prove both existence and uniqueness for every split. $\square$

The diameter in this refinement is the smallest one allowed by the matching threshold. Larger diameters already give the strict bound $L(T) \geq 2r + 2$ in the original problem. Thus the refinement describes the entire class in which the subtle spectral comparison is needed, including every competing central split. No bound on the number of leaves or the number of branch vertices is imposed.

References

- [1] H. Lin and D. Zhao. *Comparison between the first Steklov eigenvalue and algebraic connectivity on trees*. arXiv:2508.13466v1, Conjecture 1.3, Definition 2.4 and Lemma 2.5.